

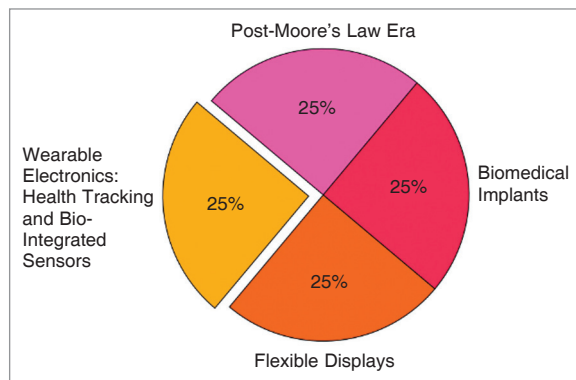
traditional silicon, are inherently flexible and stretchable. These materials are typically polymers, organic molecules, or even some forms of hybrid inorganic-organic compounds. They can be moulded, bent, and stretched without losing their electrical conductivity or functionality, making them ideal for applications where traditional, rigid semiconductors fall short.

The key to soft semiconductors lies in their molecular structure. Unlike crystalline silicon, soft semiconductors are often made up of amorphous or semi-crystalline materials, which allow them to deform without breaking. This gives them a distinct advantage in applications requiring movement, such as wearable devices, health-monitoring implants, and flexible electronics.

Why now? The emerging need for soft semiconductors

Several technological trends are converging, driving the need for new classes of semiconducting materials:

Wearable electronics. Health tracking devices and bio-integrated sensors require materials that conform to the body's natural movements. Rigid electronics offer limited comfort and flexibility, whereas soft semiconductors seamlessly integrate into wearables, such as smart clothing or skin patches.



Distribution of various applications of soft semiconductors

The science behind flexibility

One of the most exciting aspects of soft semiconductors is the transition from inorganic to organic and hybrid materials for enhanced functionality. Several categories of materials fall under the umbrella of soft semiconductors, each offering distinct advantages:

- **Organic semiconductors.** Organic semiconductors, made from carbon-based molecules or polymers, offer unique properties that allow easy manipulation and integration with various surfaces. For example, organic polymers like polythiophene can be chemically tuned to conduct electricity and are also naturally flexible, making them an excellent foundation for soft semiconductor applications.
- **Hybrid materials.** Hybrid materials like perovskites and quantum dot composites combine the strengths of organic and inorganic elements to offer improved stability and efficiency. They combine the flexibility of organics with the high performance often associated with inorganic semiconductors.
- **Elastomer-based conductors.** Made from polymers that can conduct electricity and stretch, such as PDMS (polydimethylsiloxane) and silicone, these materials are commonly used in applications requiring both resilience and electronic functionality.

Flexible displays. Companies are experimenting with rollable and foldable screens, which demand materials that remain functional under repeated mechanical stress. Traditional semiconductors, known for their brittleness, fail to meet these durability requirements.

Biomedical implants. Soft semiconductors pave the way for more effective neural implants and prosthetics. Their mechanical compatibility with biological tissues minimises immune responses and improves device longevity inside the human body.

Post-Moore's Law era. As the miniaturisation of silicon-based transistors approaches its physical limits, researchers are exploring alternative technologies, such as soft semiconductors, for innovative computing paradigms.

Recent advancements in organic chemistry, nanoscale engineering, and materials science have driven the development of soft semiconductors, a new class of flexible and stretchable electronics. Key technologies underpinning

these advancements include:

Organic field-effect transistors (OFETs). These transistors replace traditional silicon channels with organic molecules, creating a thinner and more flexible structure than conventional options. Due to their thin and pliable nature, OFETs are especially suitable for thin-film electronics and stretchable device applications.

Polymer semiconductors. Certain polymers, such as PEDOT: PSS (poly(3,4-ethylenedioxythiophene) polystyrene sulfonate), provide excellent conductivity while retaining softness and elasticity. These polymers are widely used in applications requiring flexible sensors and electrodes, combining electrical performance with physical adaptability.

Hybrid organic-inorganic systems. By blending organic semiconductors with materials like perovskites, engineers have achieved enhanced performance without losing flexibility. These hybrid systems play a significant role in developing soft solar cells and flexible optoelectronic devices, where both efficiency and adaptability are critical.

Nano-patterning and 2D materials. Ultrathin materials